

STAINLESS STEEL SINGLE POINT LOAD CELL FOR HARSH ENVIRONMENTS

capacities 15kg - 400kg



The T12 double bending beam, stainless steel single point load cell is ideal for high accuracy weight measurement with off-centre loads in harsh industrial environments. Its hermetically sealed and fully welded construction, with protection class IP68 and IP69K, is essential for use in applications involving regular wash down at high temperatures and pressures, such as the food and pharmaceutical industries. It is approved to 3000 divisions OIML R60 Class C (15-350 kg) and approved to 5000 divisions NTEP Class III (15-400 kg).

Typical applications include weighing platforms with dimensions up to 800mm x 800mm; bagging, filling and dosing machines; medical scales; checkweighers; small bucket and hopper systems.

The Polyurethane cable provides excellent protection against chemicals, UV radiation, rodents and damage from cuts/impacts. As an option, T12 load cells are available with dual-labelled ATEX and IECEx certification for all Gas and Dust zones. The certification for ATEX and IECEx dust zones 20, 21 and 22, and for IECEx gas zone 2, does not require safety barriers, which creates a large cost saving.

For applications in aggressive environments where stress corrosion or acid attack could be a problem, the T12 can be supplied with an optional Parylene coating.

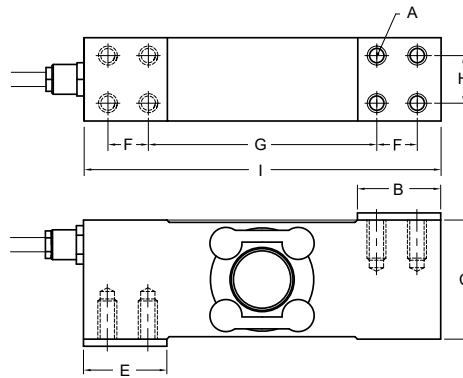
For high temperature applications, an optional version of the T12 is available for operation in environments up to 150°C (continuous temperature).

- Stainless steel construction
- 5 Year warranty
- Hermetically sealed and fully welded to IP68 and IP69K
- Options of  and  approval, or Parylene coating or high temperature version
- OIML R60 Class C 3000 divisions (C3) 15-350 kg
- Wide operating temperature range
- NTEP Class III 5000 divisions
- Polyurethane cable provides excellent protection against chemicals, UV radiation, rodents and damage from cuts/impacts
- High quality stainless steel cable entry
- High accuracy with off-centre loads



T12

technical specification...



Qty. 2 spacer plates supplied with load cell, as shown above

2 body sizes are available: see below dimensions table. The 250kg and 400kg capacity versions are larger than the other capacity versions.

T12 Load Cell

	Load cell specifications	Units
Load Cell Capacities (E_{max} or Cap)	15, 20, 30, 50, 75, 120, 200, 250, 350, 400	kg
Rated Output (S_n)	2	mV/V $\pm 0.1\%$
Accuracy Class (OIML)	3000 (See Note 1)	n.OIML
Accuracy Class (NTEP)	III (5000 div.)	—
Combined Error	$< \pm 0.017$	% S_n
Non-repeatability	$< \pm 0.01$	% S_n
Creep (30 minutes)	$< \pm 0.016$	% S_n
Minimum load cell verification Interval (V_{min})	OIML $E_{max} / 10000$ NTEP $Cap / 10000_{\Delta}$	kg
Temperature Effect on Zero Balance	$< \pm 0.002$	% $S_n / ^\circ C$
Temperature Effect on Span	$< \pm 0.0012$	% $S_n / ^\circ C$
Compensated Temperature Range	-10 to +40	$^\circ C$
Operating Temperature Range	-20 to +70	$^\circ C$
Minimum Dead Load	0	% S_n
Safe Overload	150	% S_n **
Ultimate Overload	200	% S_n **
Zero Balance	$< \pm 2$	% S_n
Maximum Deflection at Nominal Capacity	0.3-0.5	mm
Input Resistance	400 ± 20	Ω
Output Resistance	350 ± 3	Ω
Insulation Resistance	> 5000	M Ω @ 100V
Recommended Supply Voltage	5-15	V
Maximum Supply Voltage	15	V
Ingress Protection Rating	IP68 and IP69K	—
Cable Length	5	m
Cable Material	Polyurethane	
Maximum Platform Size +	15, 20, 30, 50, 75, 120, 200, 350 kg 250, 400 kg	600 x 600 800 x 800
Nominal Shipping Weight	15, 20, 30, 50, 75, 120, 200, 350 kg 250, 400 kg	1.8 4.3

1. OIML C3 (3000 divisions) approval is available for capacities from 15kg to 350kg inclusive. Δ for 15kg capacity, $V_{min} = Cap/15000$ as per NTEP Class III

** Only applies to central loads on the load cell. Not for off-centre loads.

+ The load cell must be placed centrally under the platform.

Dimensions

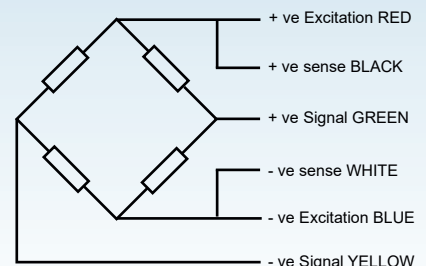
Capacity (kg)	15, 20, 30, 50, 75, 120, 200, 350	250, 400
A	8 off M8 x 1.25 x 14	8 off M10 x 1.5 x 20
B	35	50
C	50	60
D	56	66
E	35	50
F	17	30
G	96	100
H	20	40
I	150	180
J	35	60

Dimensions in mm

Model T12 ATEX and IECEx Certification

(see certificate on website for key parameter details)

Certification Type	EX Marking	Zone Application
ATEX	II 1 GD Ex ia IIC T4...T6 Ga Ex ia IIIC T85°C Da Ex ta IIIC T85°C Da	Gas zones 0, 1, 2 with safety barriers (Ex ia) Dust zones 20, 21, 22 without safety barriers (Ex ta)
IECEx	Ex ia IIC T4...T6 Ga Ex ia IIIC T200 85°C Da Ex ta IIIC T200 85°C Da (-20°C $\leq$ Tamb $\leq$ 40°C) Ex ta IIIC T200 105°C Da (-20°C $\leq$ Tamb $\geq$ 60°C) Ex ec IIC T6 Gc Ex tc IIIC T85°C Dc (-20°C $\leq$ Tamb $\leq$ 60°C)	Gas zones 0, 1, 2 with safety barriers (Ex ia) Dust zones 20, 21, 22 with safety barriers (Ex ia) Dust zones 20, 21, 22 without safety barriers (Ex ta) Gas zone 2 without safety barriers (Ex ec) Dust zone 22 without safety barriers (Ex tc)



Electrical Connections

Via 6 wire, 5.7mm diameter, screened polyurethane cable. Screen not connected electrically to load cell.

DISTRIBUTED BY:



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Our policy is one of continuous product enhancement. We therefore reserve the right to incorporate technical modifications without prior notification.



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